



Screen-Based vs. Non-Screen Leisure: How Social Context Shapes Momentary Happiness and Stress

Farangiz Akhadova '27, Data Science Major Capstone

Introduction

Leisure activities make up an important part of everyday life, but not all leisure may relate to well-being in the same way. Screen-based leisure, such as watching television or using a computer for fun, is often treated as interchangeable with other free-time activities, even though it may differ from non-screen leisure like reading, and arts and crafts. At the same time, social context may matter just as much as activity type. Doing an activity alone may feel different from doing it with other people. This project asks whether momentary happiness and stress differ across these kinds of leisure episodes, and whether those differences depend on whether a person is alone or with others.

Research Question

How do momentary happiness and stress differ between screen-based and non-screen leisure activities, and do those patterns change depending on whether people are alone or with others?

Data

This project uses the American Time Use Survey (ATUS) Well-Being Module for 2010, 2012, and 2013. The ATUS collects information on how people in the United States spend their time, and the Well-Being Module records momentary emotional ratings for selected activity episodes. I combined ATUS activity, well-being, respondent, roster, and Who files to create an episode-level dataset. The final analytic sample contains 12,444 respondents and 14,986 leisure activity episodes. I excluded episodes with invalid happiness or stress values and kept only episodes classified as either screen-based or non-screen leisure.

Variables / Sample Description

Key Variables

The two outcomes are momentary happiness and momentary stress, both measured on a 0–6 scale. The main predictors are activity type (screen-based vs. non-screen leisure) and social context (alone vs. with others). Screen-based leisure includes television, leisure computer use, and games, while non-screen leisure includes reading, arts and crafts, and relaxing. I also included age, sex, and employment status as control variables in the regression models.

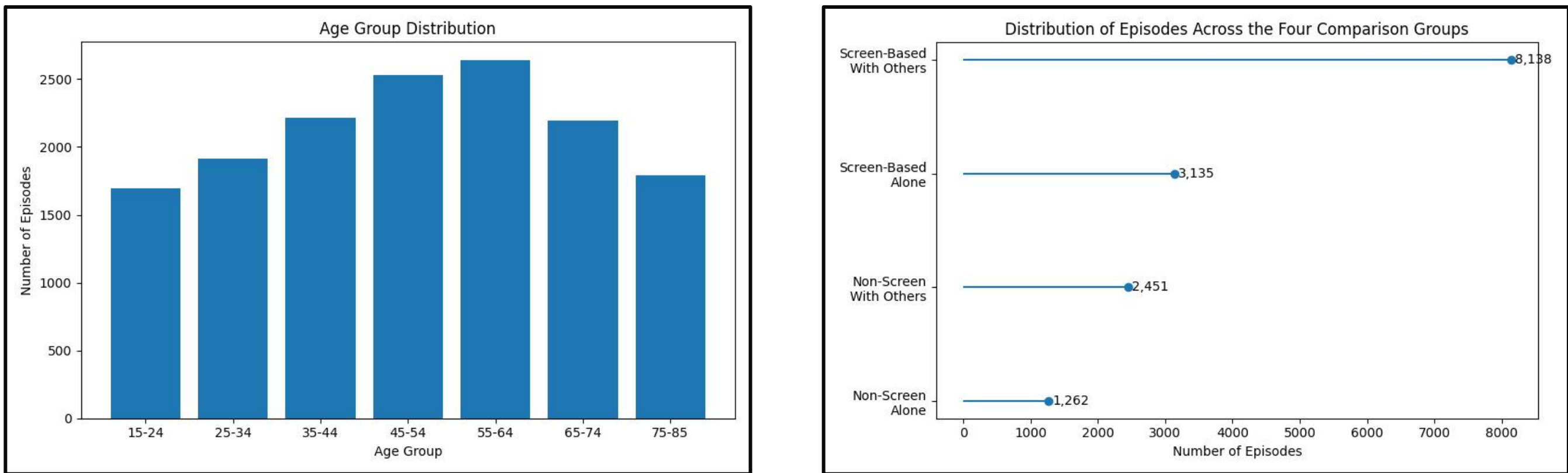
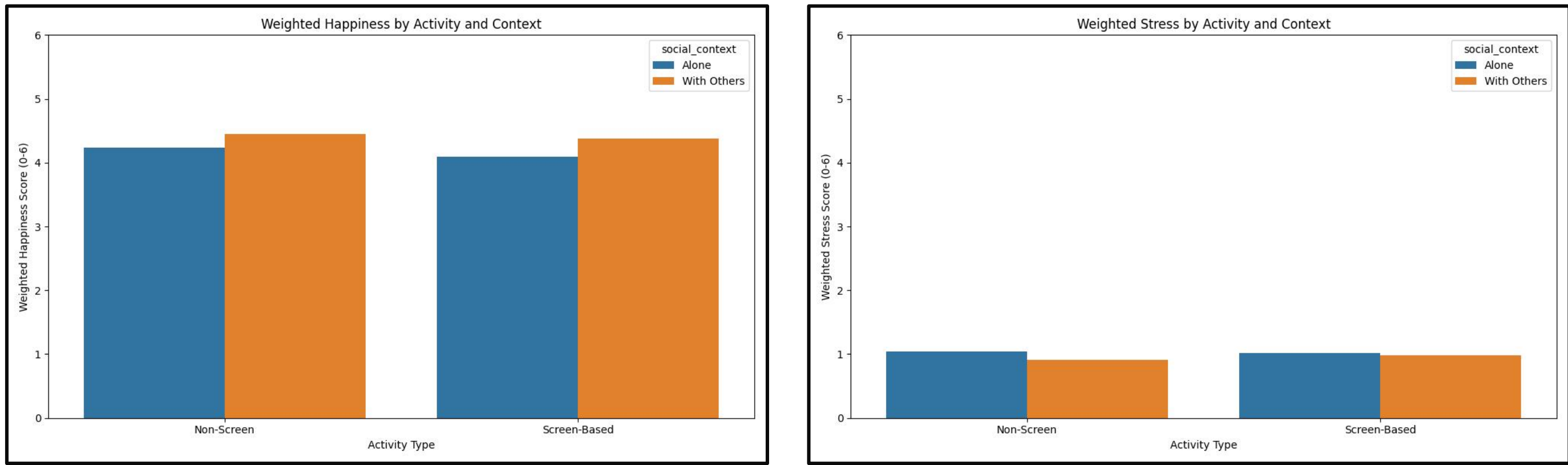
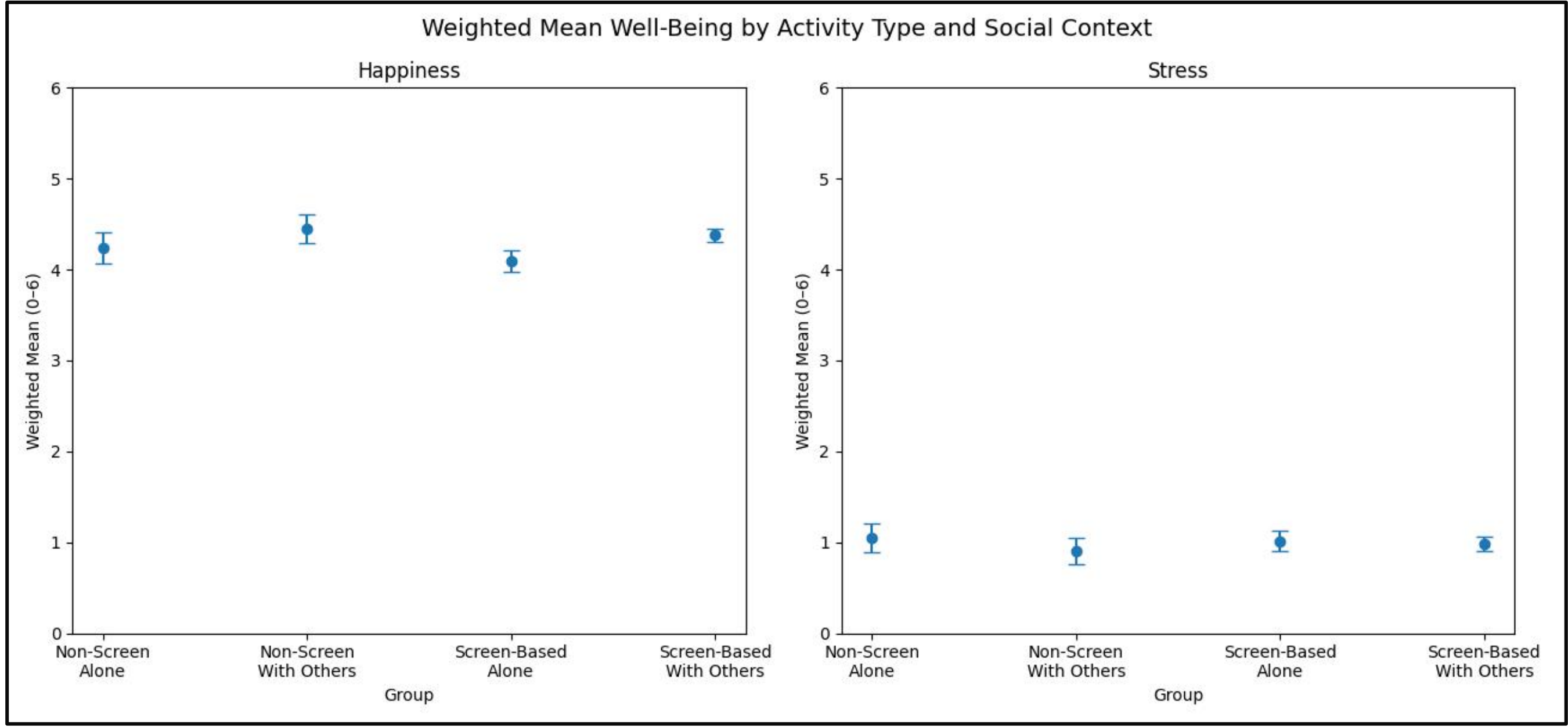
Sample Description

The final sample includes more screen-based than non-screen episodes overall. Among the four comparison groups, the largest group was screen-based leisure with others (8,138 episodes) and the smallest was non-screen leisure alone (1,262 episodes). The sample had a mean age of about 50.3 years, and the analysis also includes variation in employment status and sex.

Methods

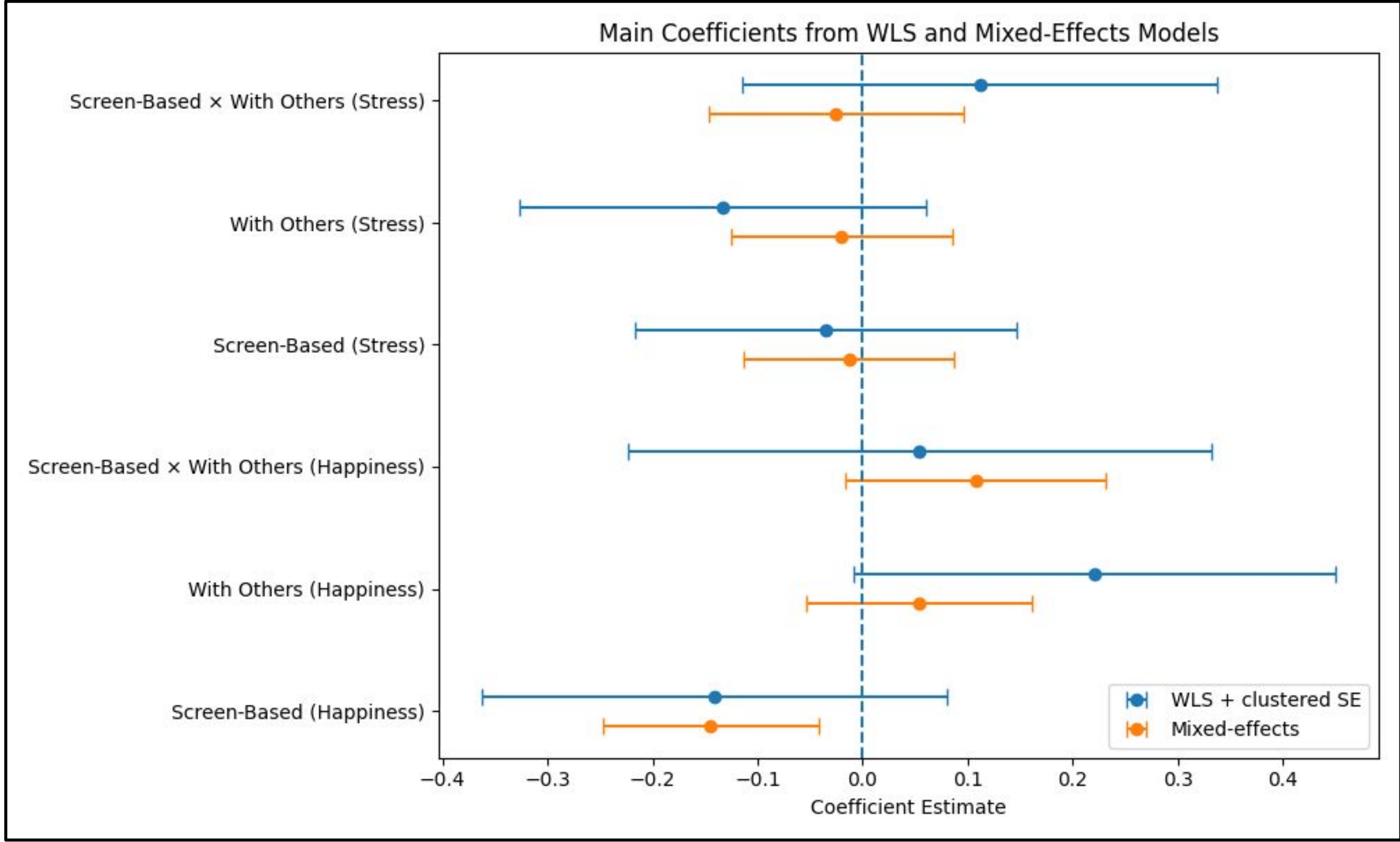
First, I merged the ATUS activity, well-being, respondent, roster, and Who files using respondent and activity identifiers. Next, I created a six-digit activity code and grouped leisure episodes into two categories: screen-based and non-screen leisure. I also created a social context variable indicating whether the respondent was alone or with others during the episode.

I then calculated weighted mean happiness and weighted mean stress for the four activity-by-context groups using the ATUS activity weight. Because the final dataset contains 14,986 episodes but 12,444 respondents, and 2,360 respondents contributed more than one episode, the observations were not fully independent. To account for that repeated-episode structure, I estimated two types of regression models for each outcome: (1) weighted least squares (WLS) with clustered standard errors by respondent, and (2) mixed-effects models with a random intercept for respondent. Both model types included activity type, social context, their interaction, and controls for age, sex, and employment status. Using both models made it possible to check whether the main findings were consistent across different approaches to within-respondent dependence.



Result	WLS + clustered SE	Mixed-effects model
Screen-Based (Happiness)	-0.141 (p = .212)	-0.144** (p = .006)
With Others (Happiness)	+0.221 (p = .060)	+0.054 (p = .327)
Screen-Based × With Others (Happiness)	+0.054 (p = .701)	+0.108 (p = .087)
Screen-Based (Stress)	-0.035 (p = .708)	-0.013 (p = .800)
With Others (Stress)	-0.133 (p = .177)	-0.020 (p = .708)
Screen-Based × With Others (Stress)	+0.112 (p = .332)	-0.025 (p = .682)

Table 1: Model Results Table



Results

Descriptive Results

Weighted happiness was highest for non-screen leisure with others (4.768) and lowest for screen-based leisure alone (4.090). Weighted stress was highest for non-screen leisure alone (1.047) and lowest for non-screen leisure with others (0.908). Overall, happiness showed a clearer difference across groups than stress.

Model Results

The regression results showed a clearer pattern for happiness than for stress. In the happiness models, the clearest result was for screen-based leisure. In the mixed-effects model, screen-based leisure was associated with lower happiness than non-screen leisure (-0.144, p = .006). In the WLS happiness model, being with others was associated with higher happiness (+0.221), but that result was only marginal (p = .060). The happiness interaction was positive in both models, which suggests that being with others may slightly lessen the happiness gap for screen-based leisure, but the interaction was not statistically significant in either model.

For stress, the main predictors were weak and inconsistent. In the WLS model, being with others was associated with lower stress (-0.133), and screen-based leisure was also associated with slightly lower stress (-0.035), but neither result was statistically significant. The interaction term was positive in the WLS stress model (+0.112), which would imply slightly higher stress for screen-based leisure with others relative to what the main effects alone would suggest, but again this was not significant. The mixed-effects stress model showed the same overall pattern of weak results.

The two model types pointed to a similar broad conclusion: the results were clearer for happiness than for stress. Among the controls, employment status was the most consistent. Respondents who were unemployed and looking for work (TELF5 = 4) or on temporary layoff (TELF5 = 5) tended to report lower happiness and higher stress than the reference group. Female respondents (TESEX = 2) also showed slightly higher predicted happiness and stress in the mixed-effects models, while older respondents reported slightly lower stress.

Outcome	Model	Significant controls
Happiness	WLS	Employment status (TELF5 = 5): -0.208 (p = .002)
Happiness	Mixed-effects	Sex (TESEX = 2): +0.099 (p = .001); Employment (TELF5 = 4): -0.136 (p = .034); Employment (TELF5 = 5): -0.122 (p < .001)
Stress	WLS	Employment (TELF5 = 4): +0.347 (p = .006); Employment (TELF5 = 5): +0.230 (p < .001)
Stress	Mixed-effects	Sex (TESEX = 2): +0.080 (p = .007); Employment (TELF5 = 4): +0.308 (p < .001); Employment (TELF5 = 5): +0.168 (p < .001); Age: -0.002 (p = .019)

Table 2: Significant Control Results

Model Fit

The WLS models had low adjusted R-squared values, suggesting that the included predictors explained only a modest share of episode-level variation. The mixed-effects models had low marginal R-squared but much larger conditional R-squared values, indicating that respondent-level differences explained a substantial share of the variation. Even so, both approaches pointed to the same overall conclusion: the results were more consistent for happiness than for stress.

Model	Outcome	Main fit statistic
WLS	Happiness	Adjusted R-squared = 0.009
WLS	Stress	Adjusted R-squared = 0.007
Mixed-effects	Happiness	Marginal R-squared = 0.016; Conditional R-squared = 0.575
Mixed-effects	Stress	Marginal R-squared = 0.004; Conditional R-squared = 0.636

Table 3: Summary of Model Fit and Consistency

Conclusion

Momentary happiness differed more clearly across leisure contexts than momentary stress. In the descriptive results, non-screen leisure with others had the highest happiness, while screen-based leisure alone had the lowest. In the regression models, the clearest finding was that screen-based leisure was associated with lower happiness in the mixed-effects model. By contrast, the stress results were small and inconsistent across models. Taken together, these findings suggest that differences in leisure-related well-being appeared more clearly for happiness than for stress.

Limitations

This project uses observational data, so the results should not be interpreted as causal. The well-being measures were recorded only for selected activity episodes, the findings depend partly on how leisure activities were grouped, and happiness and stress were self-reported at a single moment in time. Social context was also simplified to alone versus with others

References

U.S. Bureau of Labor Statistics, 2010–2013, *American Time Use Survey (ATUS) Well-Being Module Multi-year Microdata Files*, [data set], U.S. Department of Labor, bls.gov.
U.S. Bureau of Labor Statistics, 2010–2013, *ATUS Well-Being Module Data Dictionary*, [documentation], U.S. Department of Labor, bls.gov.
Fleche, S., Smith, C., July 2017, *Time Use Surveys and Experienced Well-being in France and the United States* (working paper), OECD Statistics Working Papers, No. 2017/07, OECD Publishing, Paris.